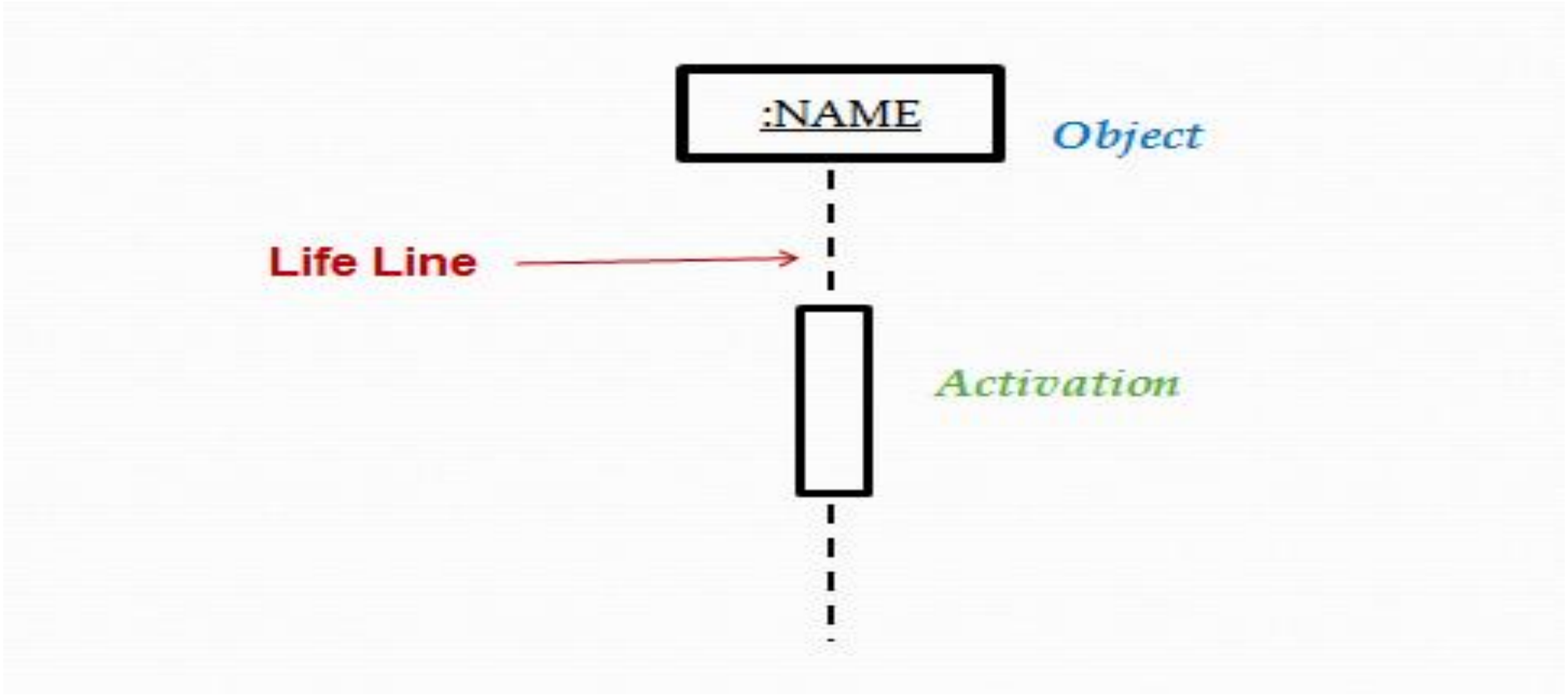
Sequence Diagram

Sequence Diagram

- Precise description of activities.
- Suitable as means for design and implementing communications.

- The **sequence diagram consists of**
  - **Objects** represented in the usual way (as named rectangles with the name underlined)
  - **Messages** represented as solid-line arrows
  - **Time** represented as a vertical progression.

2-Sequence Diagram



3- Messages

- There are two types of messages

1. **Call (synchronous) message**

   - This is a request from the object sending the message to another object that receives the message.
   - Usually, this entails the **sender waiting for the receiver** to carry out that operation.





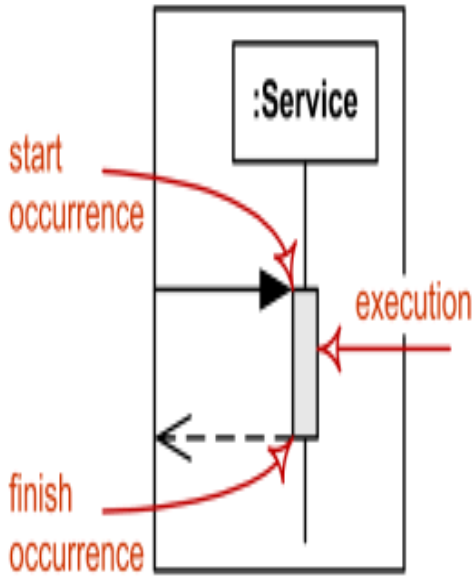
2. **Asynchronous message**

   Sender **transfers control to the receiver** and **doesn't wait for the operation to complete.**



4-Time

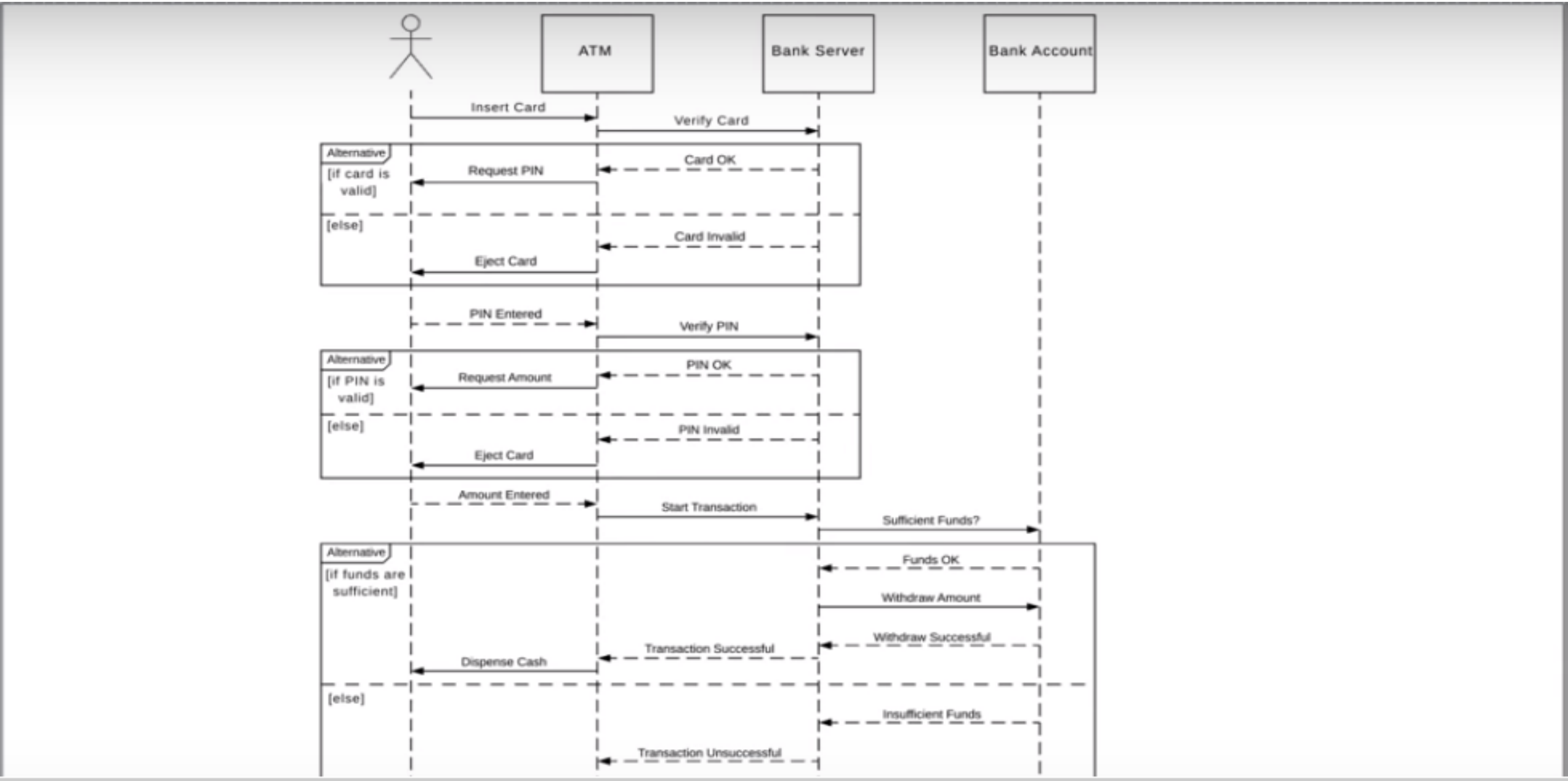
- **Execution** (full name - **execution specification**, informally called **activation**) is **interaction fragment** which represents a period in the participant's lifetime when it is
- executing a unit of behavior or action within the **lifeline**,
- sending a signal to another participant,
- waiting for a reply message from another participant.
- The **duration** of an execution is represented by two
- **execution occurrences** - the **start** occurrence and the
- **finish** occurrence.
- **Execution** is represented as a thin grey or white rectangle
- on the lifeline.



Duration of an execution is represented by two execution occurrences - start and finish.

Sequence for ATM

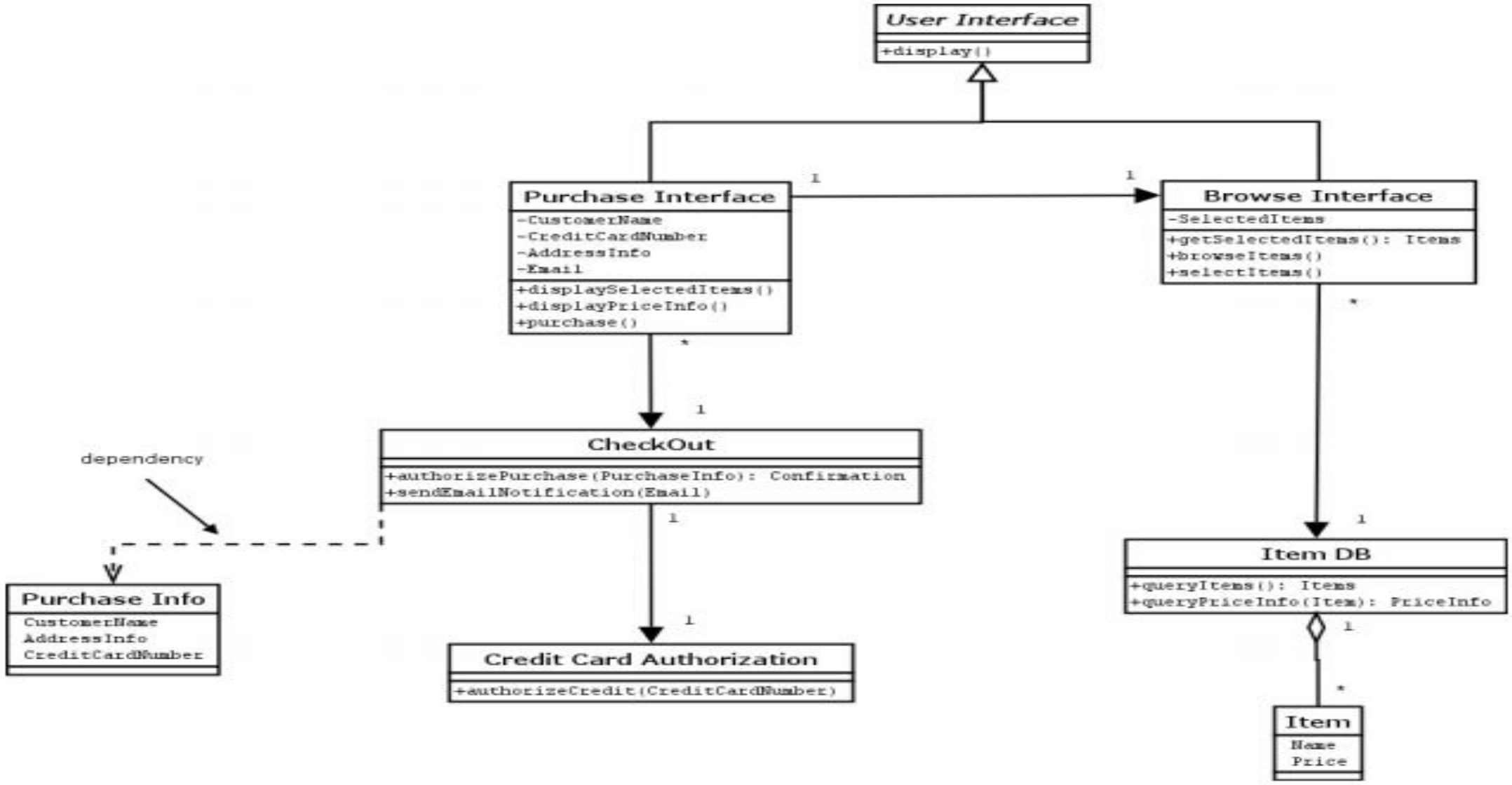
Sequence for ATM



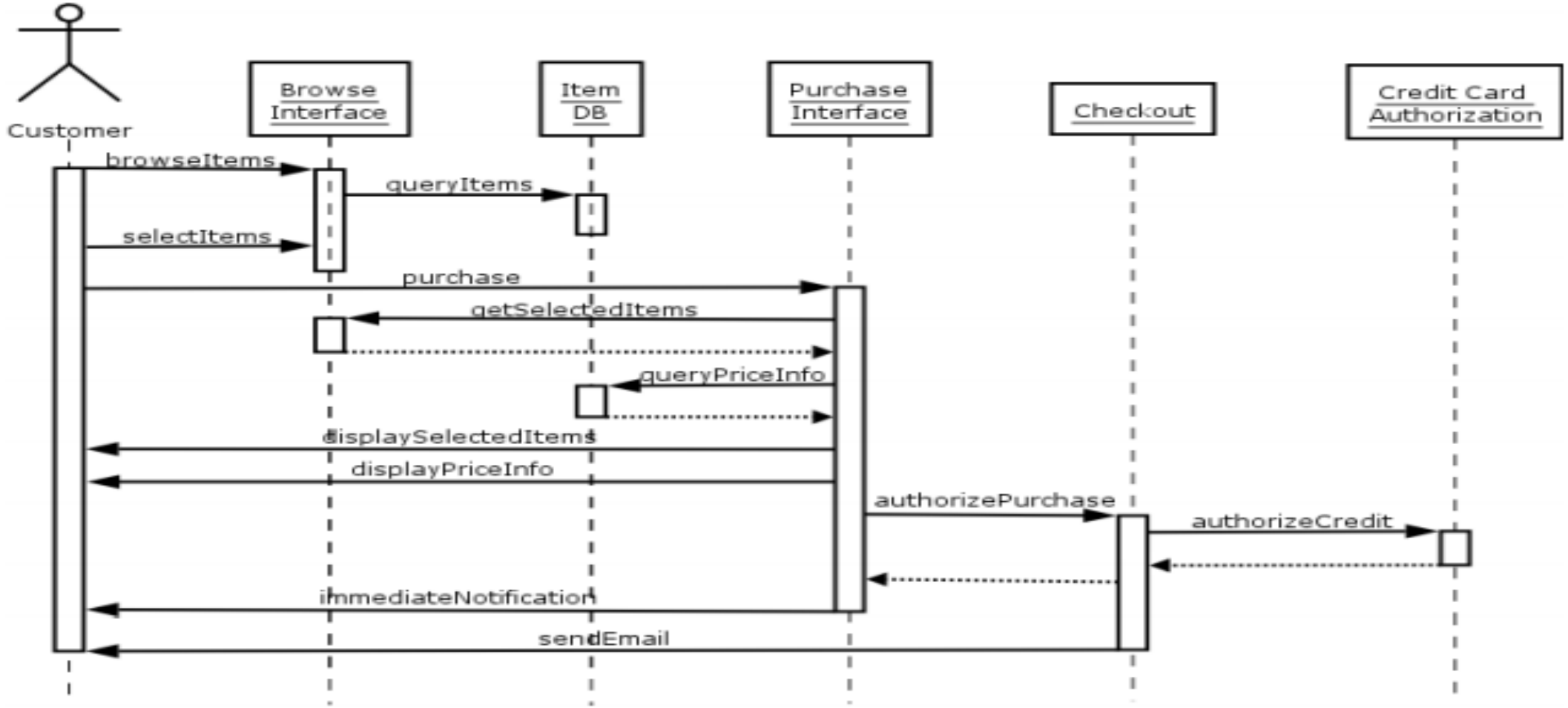
Example: Online Shopping

- In order to develop our sequence diagram we will need the use case text and the specification class diagram: Use Case Text: Buy a Product Online
  - 1. Customer browses through catalog and selects items to buy.
  - 2. Customer goes to checkout.
  - 3. Customer fills out shipping information.
  - 4. System presents full pricing information, including shipping information.
  - 5. Customer fills in credit card information.
  - 6. System authorizes purchase.
  - 7. System confirms sale immediately.
  - 8. System sends confirming email to customer. The specification class diagram as developed previously:

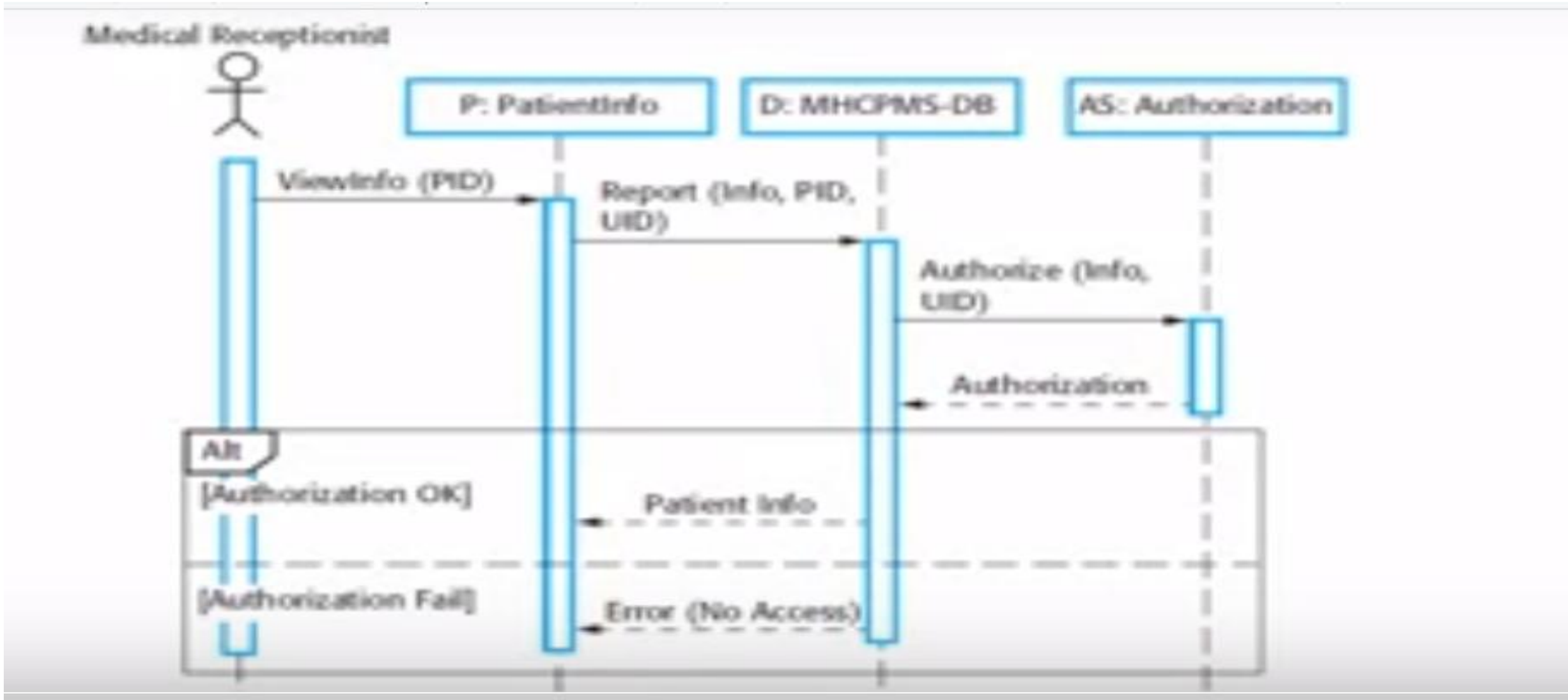
Class Diagram



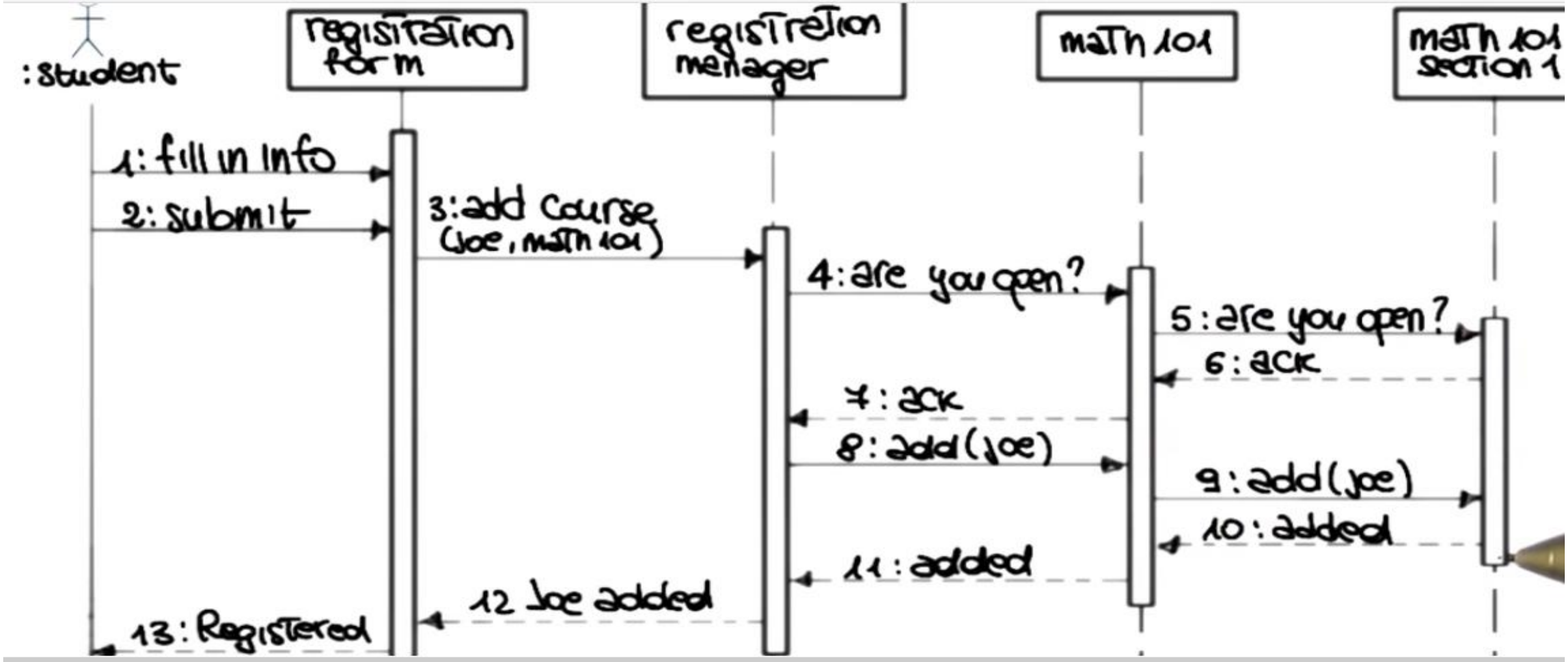
Sequence Diagram for online shopping



Sequence Diagram for Medical system



Sequence Diagram for Registration Form



What is State Diagram

- It presents the states an object can be in along with the transitions between the states and shows the starting point and endpoint of a sequence of state changes.

- Examples
    1. Switching light, its state changes from Off to On.
    2. Clicking a remote control, TV changes its state from showing you one channel to showing you another.
    3. After an appropriate amount of time, a washing machine changes its state from Washing to Rinsing.
    4. project: submitted, evaluated, accepted, rejected, monitored, audited, finished, delayed

2-Difference Between State Diagram and Other Diagrams

- The diagrams you've already studied model a **group of classes, objects, or use cases**.

- A **state diagram** shows the states of a **single object.**

State example

- **Example**

Considering the example shown in the above figure, the transition from Waiting state to Riding state takes place when the person gets a taxi. Likewise, the final state is reached, when he reaches the destination. These two occurrences can be termed as events Get_Taxi and Reach_Destination. The following figure shows the events in a state machine.



State machine Diagram example

Car Agency problem

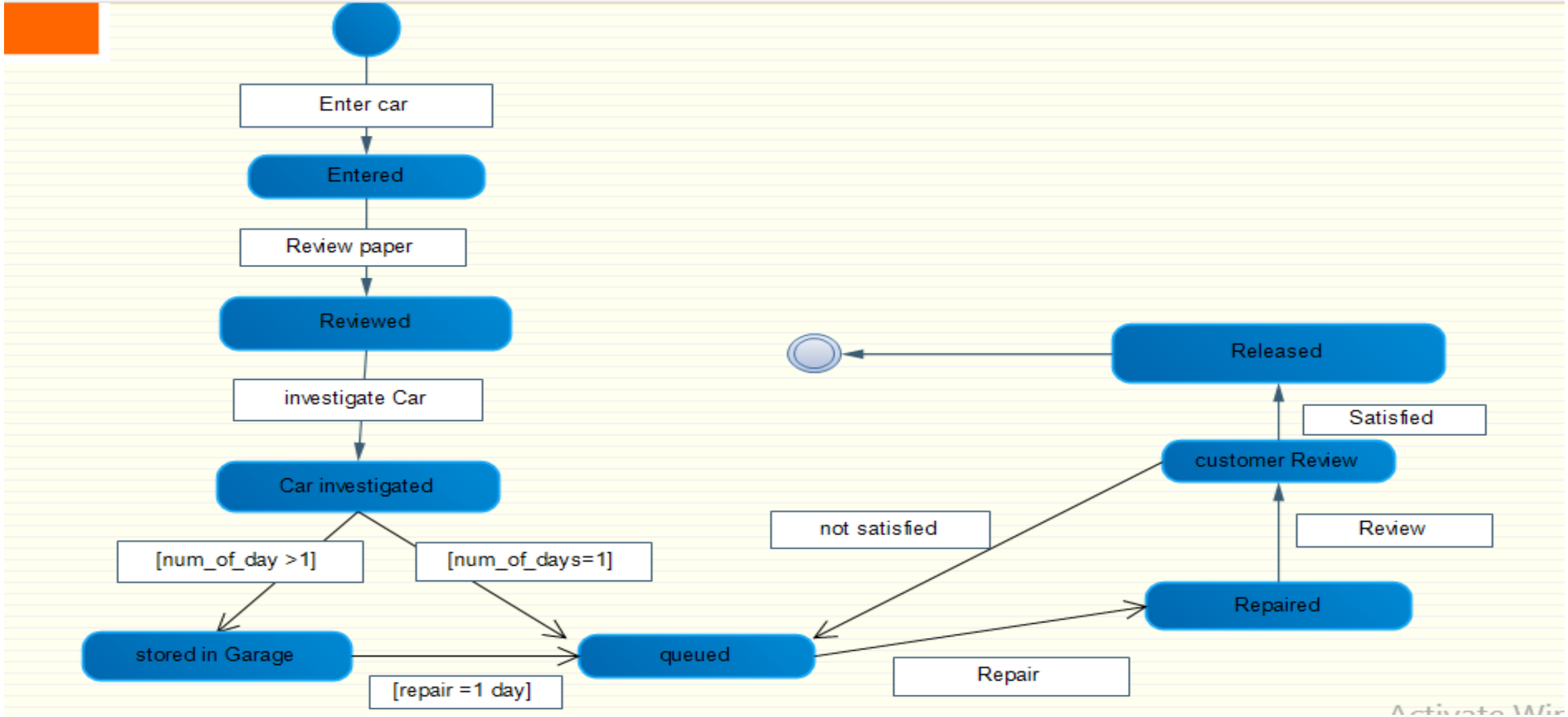
In car Agency : the customer enters the car then car’s papers reviewed after that the car is investigated if the repair take one day

So it will be queued, if it is more than one day it will be stored in Garage.

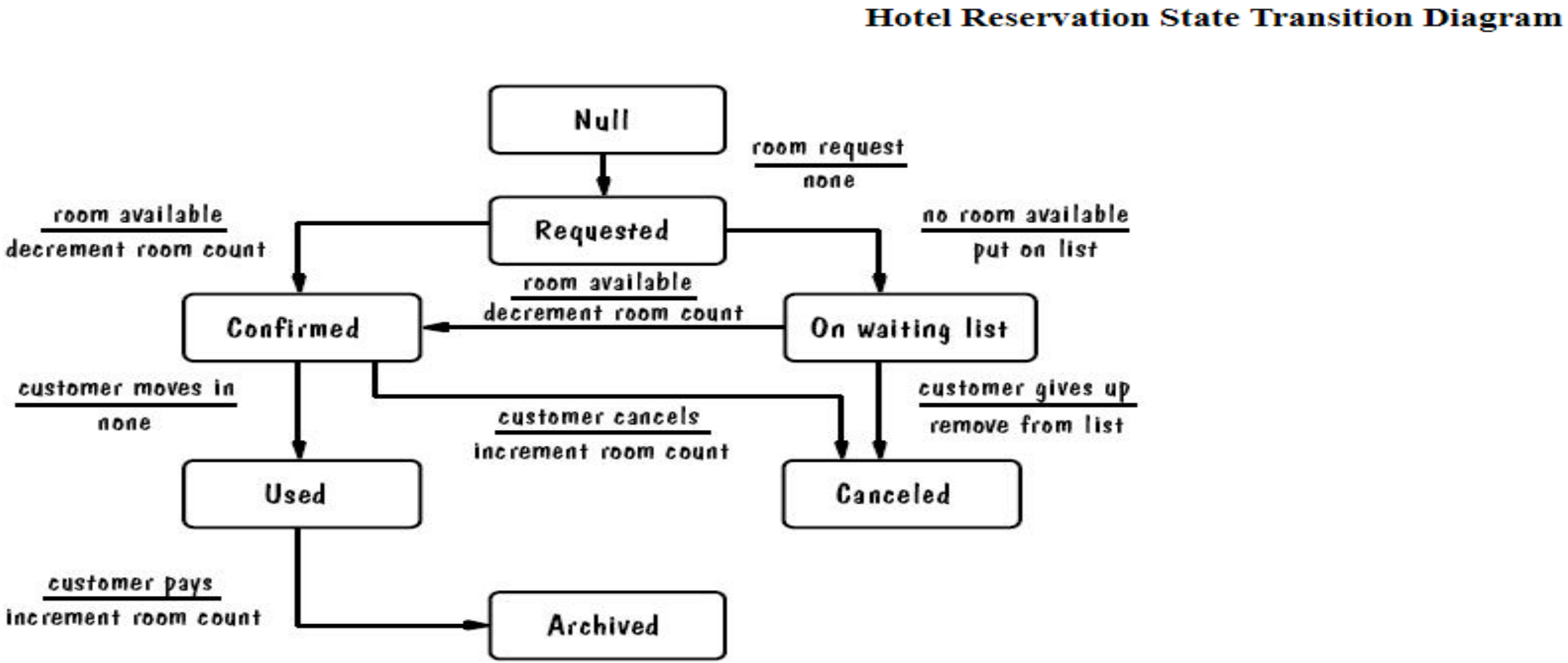
It will be moved from Garage to queue when repair day is today .

It will be moved from the queue to customer check after repaired ,if customer not satisfied it will be queued if satisfied it will be Released.

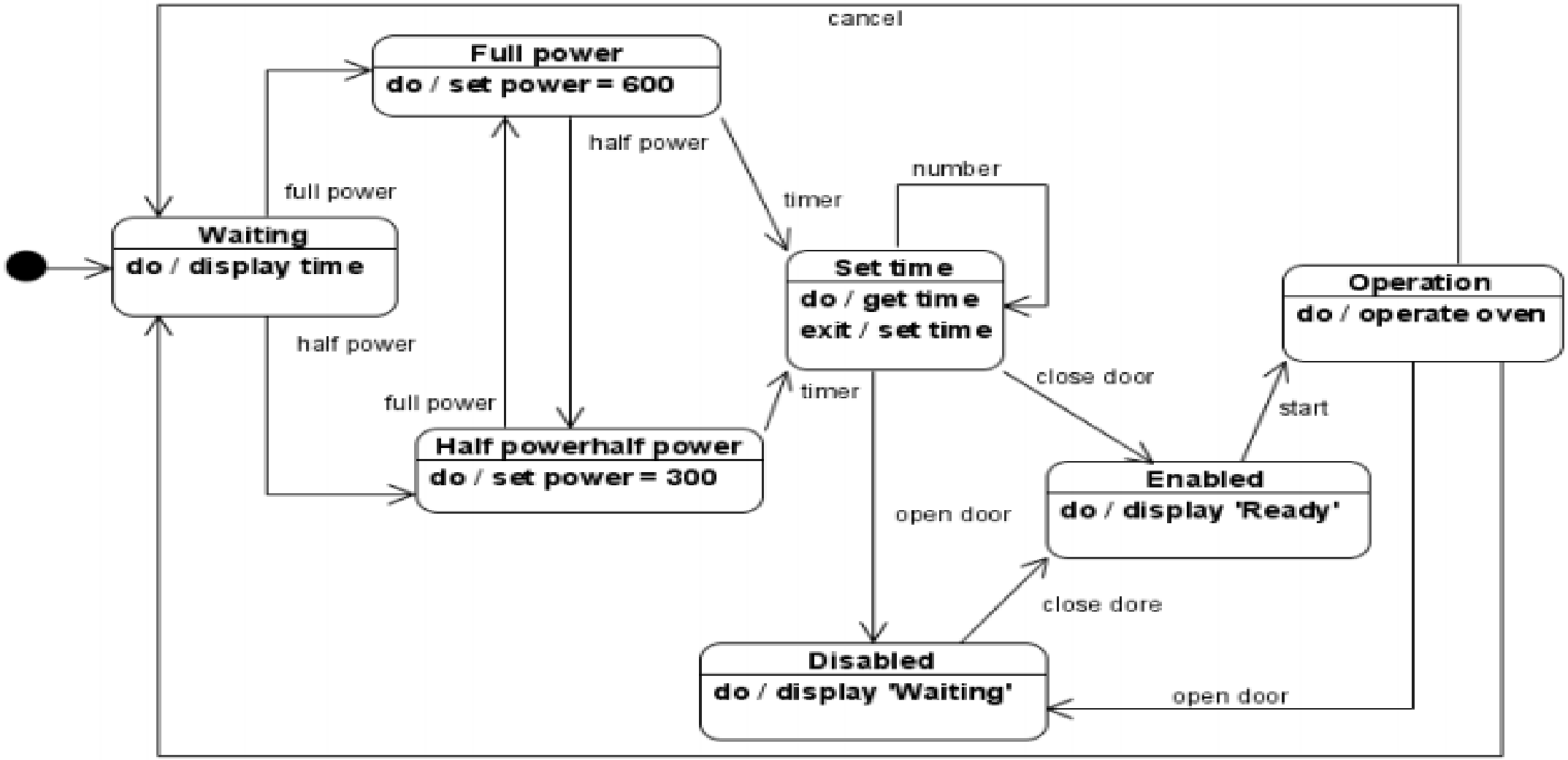
State machine Diagram for car agency



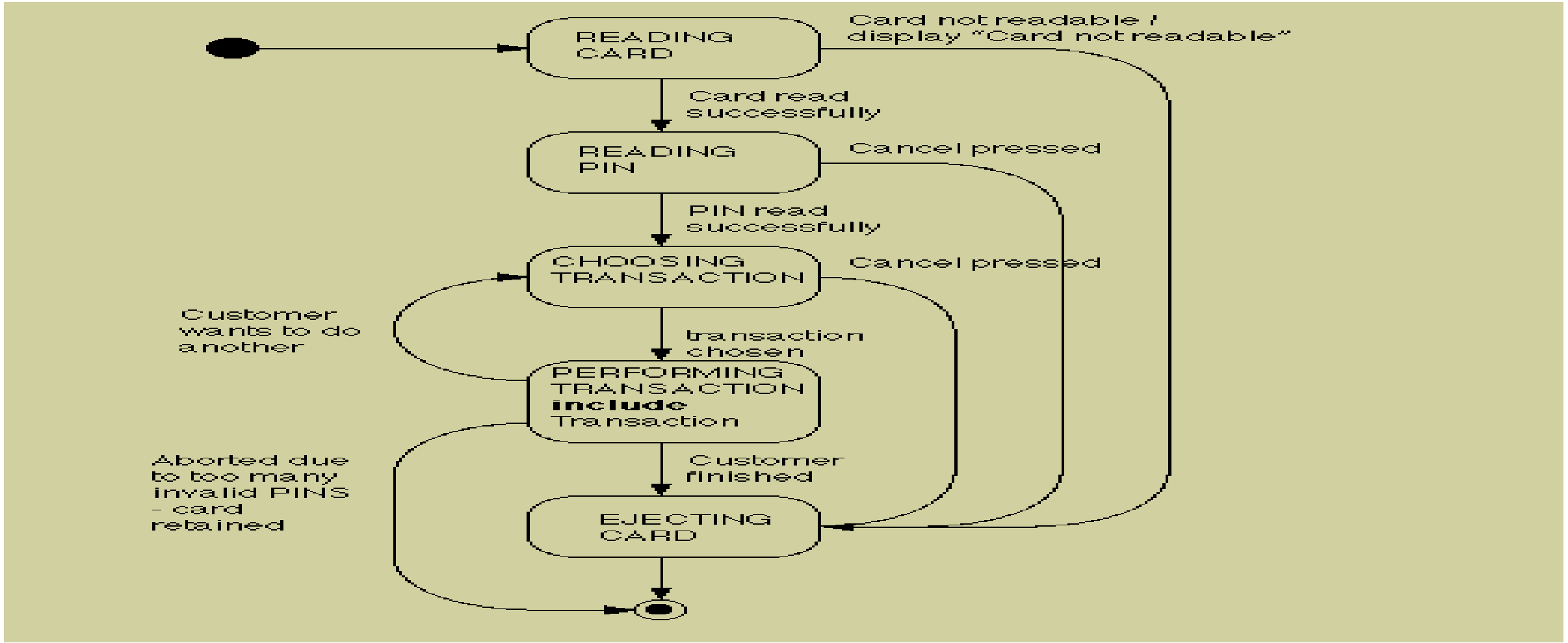
Hotel state machine diagram



State Diagram for microwave oven



State Diagram for ATM



Thank you